4	(8)	0.25	
	$\mu(0) = m_0 g / u$	0.25	**0.75**
	(9)	0.25	
5		0.5	
	(10)	0.5	
	(11)	0.5	**2.0**
	(12)	0.5	
6	(13)	0.5	
	(14)	0.25	**1.0**
	(15)	0.25	
7	(16)	0.25	
	(17)	0.5	
	(18)	0.25	**2.5**
	(19)	0.5	
	(20)	0.5	
8	(19) (22)	0.5	
	(21)	0.25	
	(22)	0.25	**1.5**
	(23)	0.5	
9	(25)	0.25	**0.25**
10	(26)	0.25	**0.25**
합계			**10,0**

문제 3 메타물질 (10점)

1. $[r, r+dr]$. $d\rho$

$$d\rho = \sigma_0 \frac{dS}{L} = \beta r \frac{2\pi r dr}{L}, \quad (1)$$

$$\rho = \int_0^R d\rho = \frac{2\pi\beta R^3}{3L}. \quad (2)$$

$$R_0 = \frac{1}{\rho} = \frac{3L}{2\pi\beta R^3} = 2.39 \times 10^{-2} Ohm. \quad (3)$$

2.

$$P_I = I^2 R_0. \quad (4)$$

, .

$$P_I = 2\pi RLP_{ext} = 2\pi\alpha RL(T_s - T_0), \quad (5)$$

$$T_s = T_0 + \frac{3I^2}{4\pi^2\alpha\beta R^4} = 297K. \quad (6)$$

3. r .

.

$$j = \sigma_0 E, \quad (7)$$